Native Complex Geometric Brownian Motion

Itô correction, stochastic rotation, and noise-dimension geometry

Complex–Itô Random-Walk Research Project

July 24, 2026

We construct a native complex exponential driven by one real Brownian motion whose modulus is exactly a prescribed geometric Brownian motion and whose imaginary log increment supplies deterministic and stochastic rotation. For real parameters $\mu, \sigma, \omega, \beta$ and $\mathcal{Z}_0 \neq 0$, the process is

$$\mathcal{Z}_t = \mathcal{Z}_0 \exp([(\mu - \sigma^2/2) + i\omega]t + (\sigma + i\beta)W_t) .$$

Its modulus satisfies $d|\mathcal{Z}_t|/|\mathcal{Z}_t| = \mu\, dt + \sigma\, dW_t$, while a continuous phase lift is $\Theta_t = \Theta_0 + \omega t + \beta W_t$. Complex Itô calculus gives the multiplicative SDE drift $\mu - \beta^2/2 + i(\omega + \sigma\beta)$; the cross term is essential. We reconcile the ordinary exponential, exact finite-step walk, stochastic logarithm, and Doléans exponential; derive transition laws, moments, and bilinear and Hermitian brackets; and prove that one real driver forces rank-one log-polar and Cartesian local covariance. Phase 3 is recovered as a constrained subfamily, whereas independent radial and angular diffusion requires two drivers. An executable notebook supplies 18 visual explanations and seeded checks of exactness, moments, covariance rank, Euler–Maruyama convergence, and quadratic variation. Complex geometric Brownian motion is established theory; the contribution is a modulus-preserving four-parameter synthesis and an explicit account of its noise-dimension boundary.

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1 Introduction

Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

turns an imaginary exponent into rotation. Geometric Brownian motion (GBM) already has an exponential representation,

$$R_t = R_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right].$$

These observations suggest a precise design question: can one add an imaginary drift and an imaginary Brownian coefficient to the same exponent so that the result rotates while its modulus remains *exactly* the prescribed GBM? We also require the definition to begin with one complex initial value and one real Brownian motion, rather than with separately evolved Cartesian coordinates or a pre-existing state process.

The answer is the direct construction

$$\boxed{\mathcal{Z}_t = \mathcal{Z}_0 \exp\left(\left[\left(\mu - \frac{1}{2}\sigma^2\right) + i\omega\right]t + (\sigma + i\beta)W_t\right).} \tag{1}$$

The four real parameters retain separate meanings:

- μ is the arithmetic drift of the modulus;
- σ is its Brownian volatility;
- ω is deterministic angular velocity; and
- β is angular Brownian volatility.

The construction is elementary to state, but its Itô dynamics are not obtained by simply reading the ordinary exponent. If

$$B = \sigma + i\beta,$$

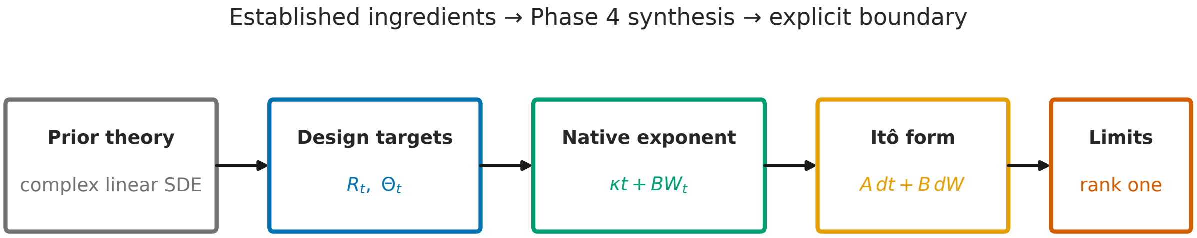

Figure 1: Flow diagram separating established complex linear SDE theory from the Phase 4 parameter synthesis and its rank-one limitation.

then $B^2 = \sigma^2 - \beta^2 + 2i\sigma\beta$. Consequently, the complex multiplicative SDE contains both a real correction $-\beta^2/2$ and an imaginary cross drift $\sigma\beta$. That square is a complex bilinear square. It must not be replaced by the nonnegative quantity $|B|^2$.

Figure 1 states the claim hierarchy used throughout this paper. Itô's formula, scalar complex linear SDEs, complex GBM, and stochastic exponentials are established (Itô 1951; Doléans-Dade 1970; Higham 2001; Buckwar et al. 2006). Our contribution is a transparent parameter synthesis, geometric interpretation, exact one-step formula, and reproducible boundary analysis. We do not claim to have invented complex GBM.

A second distinction is equally important. The informal notation $(dW_t)^2 = dt$ summarizes quadratic variation; $i^2 = -1$ is an algebraic identity. They are not the same statement. Complex multiplication organizes scale and rotation, while quadratic variation supplies the second-order term in Itô's formula. The connection is through calculus, not through an identification of the two rules.

2 Literature and positioning

Itô's original change-of-variables formula establishes the second-order correction that distinguishes stochastic from ordinary differential calculus (Itô 1951). Doléans-Dade's work places multiplicative solutions in the semimartingale exponential framework (Doléans-Dade 1970). Modern treatments make the relation between stochastic exponentials and logarithms explicit (Larsson and Ruf 2019), while semimartingale representation and multiplicative compensation frameworks include complex-valued processes (Černý and Ruf 2022, 2023).

For the constant-coefficient scalar linear equation,

$$dX_t = \lambda X_t\,dt + \eta X_t\,dW_t,$$

Higham uses the exact exponential solution as a central analytical and computational benchmark (Higham 2001). Buckwar, Horváth-Bokor, and Winkler allow the state and coefficients to be complex, keep W real, and explicitly call the exact solution "complex geometric Brownian motion" in the full text (p. 262) (Buckwar et al. 2006). Perret independently uses the same terminology for the scalar complex linear SDE with multiplicative real noise (Perret 2010). Complex scalar linear test equations also appear in numerical stability analysis (Abdulle et al. 2013), and direction-and-norm decompositions provide related geometric context for multiplicative-noise schemes (Mora et al. 2015).

The present process is therefore an established complex linear SDE under a particular parameterization. What the parameterization accomplishes is to preserve the ordinary GBM meanings of (μ, σ) in the modulus while assigning independent modeling meanings to (ω, β) in the phase. It also reveals the correction needed when translating those log-polar targets to the Itô SDE coefficients.

Classical planar Brownian motion is a different object. Its two Cartesian Brownian coordinates yield two noise directions, and its winding theory has distinct asymptotic laws (Spitzer 1958; Pitman and Yor 1986). Those references are used here only to mark the distinction. A single real W, even when multiplied by a complex number, still supplies one instantaneous random direction.

The full source and claim audit is preserved in `RESEARCH_NOTES.md`. The audit deliberately excludes claims that the complex GBM family, stochastic exponential, or exact scalar linear solution is new.

3 Preliminaries and notation

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfy the usual conditions, and let $W = (W_t)_{t\geq 0}$ be a standard **real** Brownian motion. Brownian quadratic variation is

$$[W, W]_t = t, \tag{2}$$

or differentially $d[W, W]_t = dt$. For a partition $\pi = \{0 = t_0 < \cdots < t_n = t\}$, this means

$$\sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2 \longrightarrow t$$

in probability as the mesh tends to zero; it is not an ordinary finite-increment identity (Mörters and Peres 2010).

We take

$$\mu, \sigma, \omega, \beta \in \mathbb{R}, \qquad \mathcal{Z}_0 \in \mathbb{C} \setminus \{0\},$$

and choose an initial phase lift Θ_0 satisfying $\mathcal{Z}_0 = R_0 e^{i\Theta_0}$, where $R_0 = |\mathcal{Z}_0| > 0$. Define

$$\kappa = \left(\mu - \frac{1}{2}\sigma^2\right) + i\omega, \qquad B = \sigma + i\beta. \tag{3}$$

We use two complex brackets:

$$[Z, Z]_t \quad \text{and} \quad [Z, \overline{Z}]_t.$$

The first is bilinear and can be complex. The second is Hermitian and nonnegative. Conflating them replaces B^2 by $|B|^2$ and changes the model.

"Native complex" is operational language in this paper. It means that the defining formula uses $\mathcal{Z}_0$, W_t, and complex coefficients directly. It does not claim invariance under arbitrary coordinate changes. Cartesian components may be plotted later, but they are consequences rather than primitives.

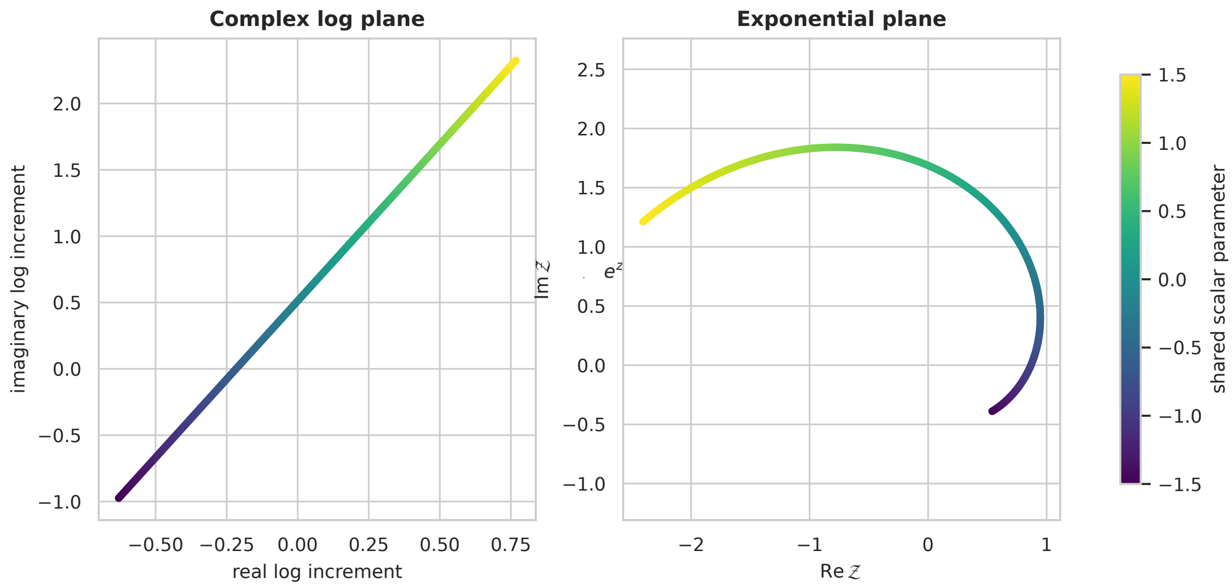

Figure 2: A colored line in the complex log plane maps through the exponential to a rotating curve with changing radius in the complex plane.

4 Problem formulation

We seek a nonzero complex process satisfying five requirements:

1. its definition contains no separately evolved X_t and Y_t;
2. its modulus is a real GBM with parameters (μ, σ);
3. its continuous phase has drift ω and Brownian coefficient β;
4. one finite Brownian increment produces an exact multiplicative step; and
5. the number of stochastic drivers remains explicit.

A Cartesian-first definition $Z_t = X_t + iY_t$ does not answer the first requirement. A post-processing map $Z_t = f(S_t)$ of a pre-existing scalar state also leaves the original coordinate system in the definition. Instead, we specify the complex log increment itself:

$$d\Lambda_t = \kappa\, dt + B\, dW_t, \qquad \Lambda_0 = 0,$$

and exponentiate:

$$\mathcal{Z}_t = \mathcal{Z}_0 e^{\Lambda_t}.$$

The two panels of Figure 2 show the organizing geometry. A single scalar Gaussian coordinate moves along an affine direction in the complex log plane. Exponentiation converts its real component to log scale and its imaginary component to phase.

5 Native complex exponential construction

5.1 Direct definition

For $t \geq 0$, define the process by Eq. 1, equivalently

$$\mathcal{Z}_t = \mathcal{Z}_0 e^{\kappa t + BW_t}. \tag{4}$$

The sample paths are continuous and adapted. Since the complex exponential never vanishes,

$$\mathcal{Z}_0 \neq 0 \quad \Longrightarrow \quad \mathcal{Z}_t \neq 0 \quad \text{for every finite } t$$

pathwise. This avoids the zero-crossing and local-time complications that arise when a signed real process is encoded literally as a polar radius.

5.2 Four-parameter separation

Taking the real and imaginary parts of the exponent yields

$$\operatorname{Re}(\kappa t + BW_t) = \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t,$$

$$\operatorname{Im}(\kappa t + BW_t) = \omega t + \beta W_t.$$

The same W_t appears in both. The parameters are freely specifiable, but the realized log-radius and phase noise are perfectly coupled whenever $\sigma\beta \neq 0$.

6 Exact modulus and continuous phase

6.1 Modulus

Using $|e^z| = e^{\operatorname{Re} z}$,

$$\boxed{R_t := |\mathcal{Z}_t| = R_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right].} \tag{5}$$

Applying real Itô calculus gives

$$\boxed{\frac{dR_t}{R_t} = \mu \, dt + \sigma \, dW_t.} \tag{6}$$

Thus the rotational parameters ω and β do not alter the modulus law. The equality is pathwise, not merely distributional.

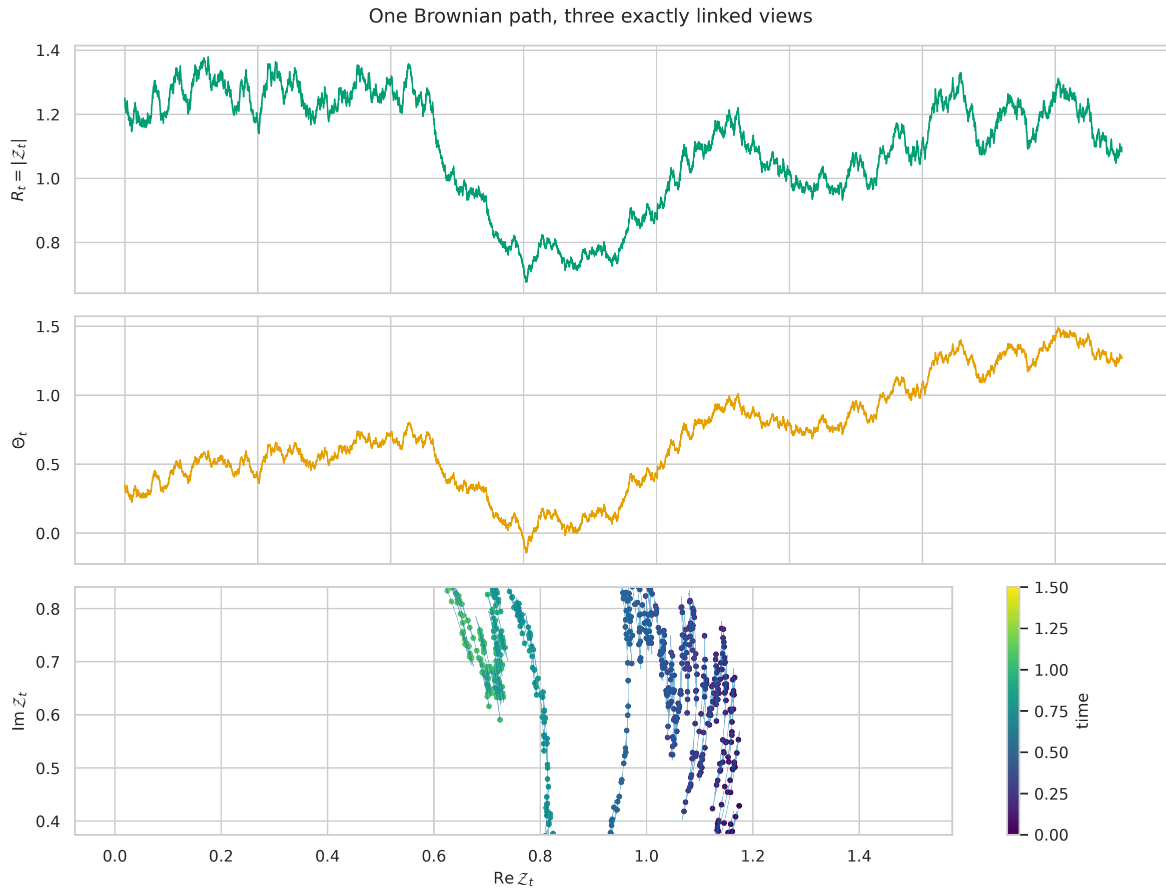

Figure 3: Three panels show the exact GBM modulus, continuous unwrapped phase, and resulting complex trajectory generated by one Brownian path.

6.2 Phase lift

Define

$$\boxed{\Theta_t = \Theta_0 + \omega t + \beta W_t.} \tag{7}$$

Then

$$\mathcal{Z}_t = R_t e^{i\Theta_t}.$$

Θ is a continuous unwrapped phase. The principal argument $\mathrm{Arg}\,\mathcal{Z}_t \in (-\pi, \pi]$ may jump by 2π at its branch cut even though $\mathcal{Z}$ and Θ are continuous. We therefore never silently identify the continuous lift with the principal branch.

Figure 3 shows three views of one Brownian realization. Neither the radius nor the phase is fixed, yet no second random input has been introduced.

The space-time view in Figure 4 prevents a common visual misinterpretation: a path drawn in a plane is not necessarily driven by two-dimensional Brownian noise.

Figure 5 varies only (ω, β) while holding the radial parameters and Brownian path fixed. Deterministic rotation changes the phase trend; stochastic rotation changes the phase's quadratic variation.

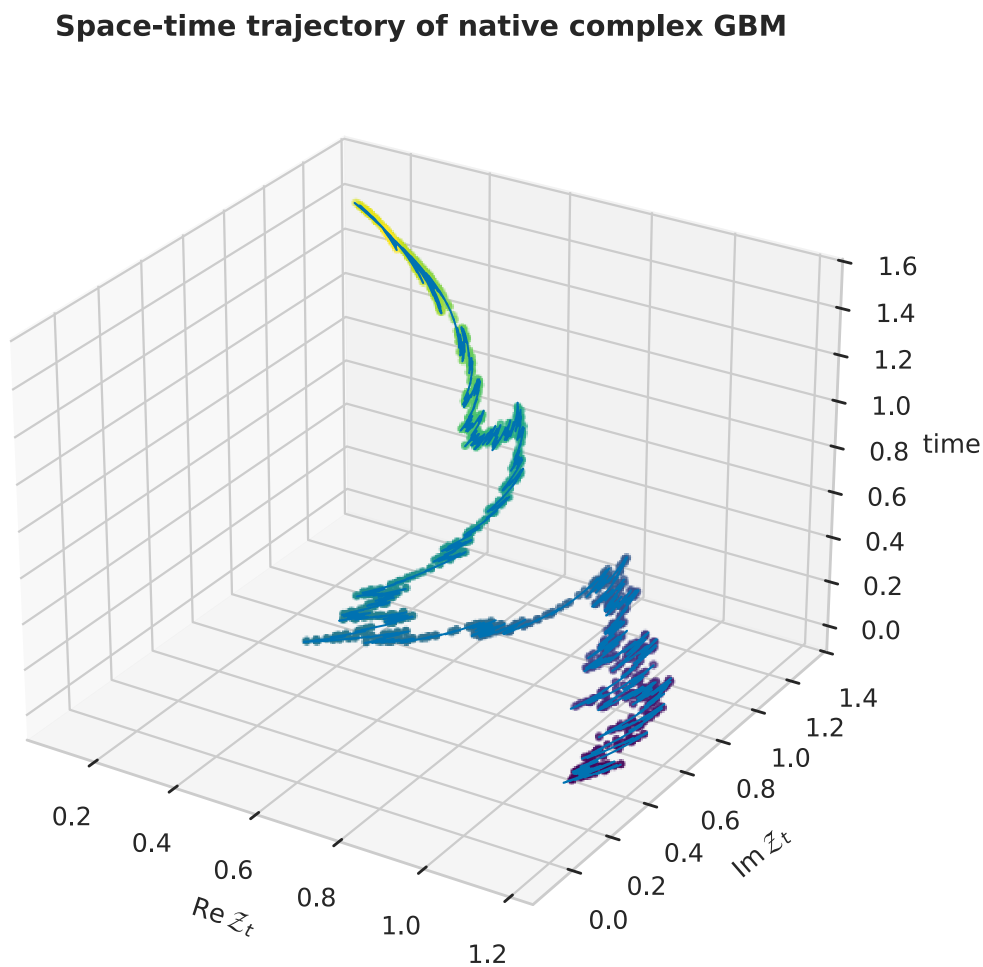

Figure 4: Three-dimensional curve showing real part, imaginary part, and time for one exact native complex geometric Brownian motion path.

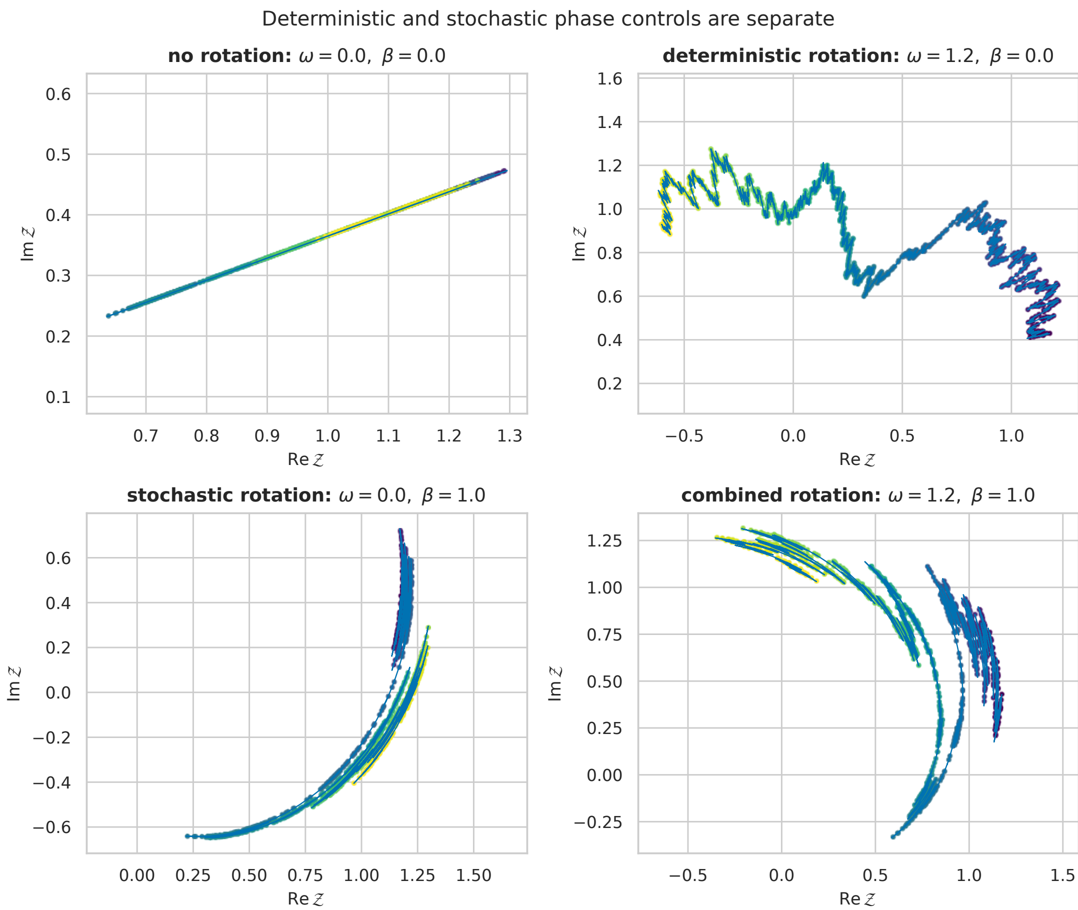

Figure 5: Four complex paths compare absent, deterministic, stochastic, and combined rotation while keeping the same radial GBM parameters and Brownian path.

7 Complex Itô dynamics

Let $F(z) = e^z$ and $\Lambda_t = \kappa t + BW_t$. The complex exponential is entire, so applying Itô's formula componentwise gives

$$d(e^{\Lambda_t}) = e^{\Lambda_t} d\Lambda_t + \frac{1}{2} e^{\Lambda_t} d[\Lambda, \Lambda]_t.$$

Because the bracket is bilinear,

$$d[\Lambda, \Lambda]_t = B^2 dt.$$

Therefore

$$\frac{d\mathcal{Z}_t}{\mathcal{Z}_t} = \left(\kappa + \frac{1}{2} B^2\right) dt + B\, dW_t. \tag{8}$$

Set $A = \kappa + B^2/2$. Expanding the square,

$$B^2 = (\sigma + i\beta)^2 = \sigma^2 - \beta^2 + 2i\sigma\beta,$$

so

$$\boxed{A = \mu - \frac{1}{2}\beta^2 + i(\omega + \sigma\beta).} \tag{9}$$

The native complex Itô SDE is

$$\boxed{\frac{d\mathcal{Z}_t}{\mathcal{Z}_t} = \left[\mu - \frac{1}{2}\beta^2 + i(\omega + \sigma\beta)\right] dt + (\sigma + i\beta) dW_t.} \tag{10}$$

Two corrections must be retained:

- $-\beta^2/2$ compensates the radial effect of stochastic rotation; and
- $+\sigma\beta$ is the imaginary cross drift caused by a shared driver.

These terms are forced by the target modulus and phase. Conversely, starting from $B = \sigma + i\beta$ and demanding $A - B^2/2 = \kappa$ determines A uniquely.

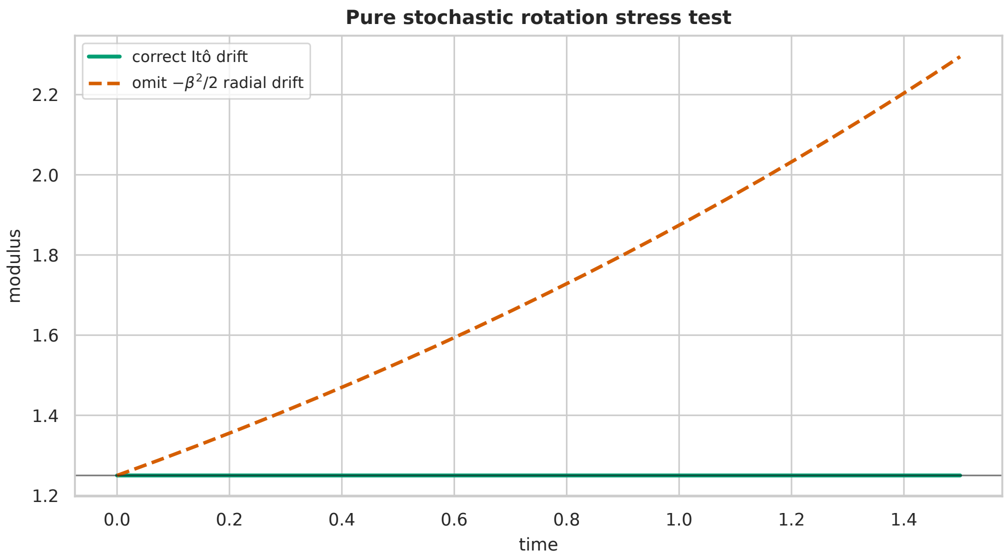

Figure 6: The correctly compensated pure stochastic rotation keeps constant modulus, while omitting the negative half beta-squared Itô drift creates spurious radial growth.

7.1 Pure stochastic rotation stress test

Take $\mu = \sigma = 0$. Then

$$\mathcal{Z}_t = \mathcal{Z}_0 e^{i\omega t + i\beta W_t}$$

has constant modulus. Its SDE nevertheless contains a real drift:

$$\frac{d\mathcal{Z}_t}{\mathcal{Z}_t} = \left(-\frac{1}{2}\beta^2 + i\omega\right) dt + i\beta\, dW_t.$$

If the term $-\beta^2/2$ is omitted from the SDE drift, the corresponding exact solution acquires the spurious radial factor $e^{\beta^2 t/2}$.

Figure 6 is a direct diagnostic of the Itô correction. Rotation generated by Brownian phase is not ordinary differentiable rotation.

8 Stochastic exponential and logarithm

Let

$$Y_t = At + BW_t.$$

For a continuous semimartingale, the Doléans exponential is

$$\mathcal{E}(Y)_t = \exp\left(Y_t - Y_0 - \frac{1}{2}[Y, Y]_t\right)$$

(Doléans-Dade 1970; Larsson and Ruf 2019). Since $[Y, Y]_t = B^2 t$,

$$\mathcal{E}(Y)_t = \exp\left[\left(A - \frac{1}{2}B^2\right)t + BW_t\right] = e^{\kappa t + BW_t}.$$

Hence

$$\boxed{\mathcal{Z}_t = \mathcal{Z}_0 \mathcal{E}(Y)_t.} \tag{11}$$

This reconciles the ordinary and stochastic exponentials. The continuous ordinary log lift and the stochastic logarithm are related but not identical:

$$\log_{\text{cont}} \frac{\mathcal{Z}_t}{\mathcal{Z}_0} = \kappa t + BW_t,$$

whereas

$$\mathcal{L}\left(\frac{\mathcal{Z}}{\mathcal{Z}_0}\right)_t := \int_0^t \frac{d\mathcal{Z}_s}{\mathcal{Z}_s} = At + BW_t.$$

Their difference is the half-bracket correction $B^2 t/2$. This is the complex analogue of the familiar distinction between the log return and the arithmetic drift of real GBM.

9 Exact transition and random-walk formula

For a grid t_n with step $h = t_{n+1} - t_n$, define $\Delta W_n = W_{t_{n+1}} - W_{t_n} \sim N(0, h)$. Independent Brownian increments give

$$\boxed{\mathcal{Z}_{n+1} = \mathcal{Z}_n \exp\left(\left[\left(\mu - \frac{1}{2}\sigma^2\right) + i\omega\right]h + (\sigma + i\beta)\Delta W_n\right).} \tag{12}$$

This is an exact random walk on the multiplicative group $\mathbb{C} \setminus \{0\}$. One complex log increment combines:

$$\Delta \log R_n = \left(\mu - \frac{1}{2}\sigma^2\right)h + \sigma \Delta W_n,$$

$$\Delta \Theta_n = \omega h + \beta \Delta W_n.$$

Figure 7 visualizes the step in the log plane and its multiplicative image. Unlike Euler–Maruyama, Eq. 12 has no time-discretization error for the constant-coefficient model.

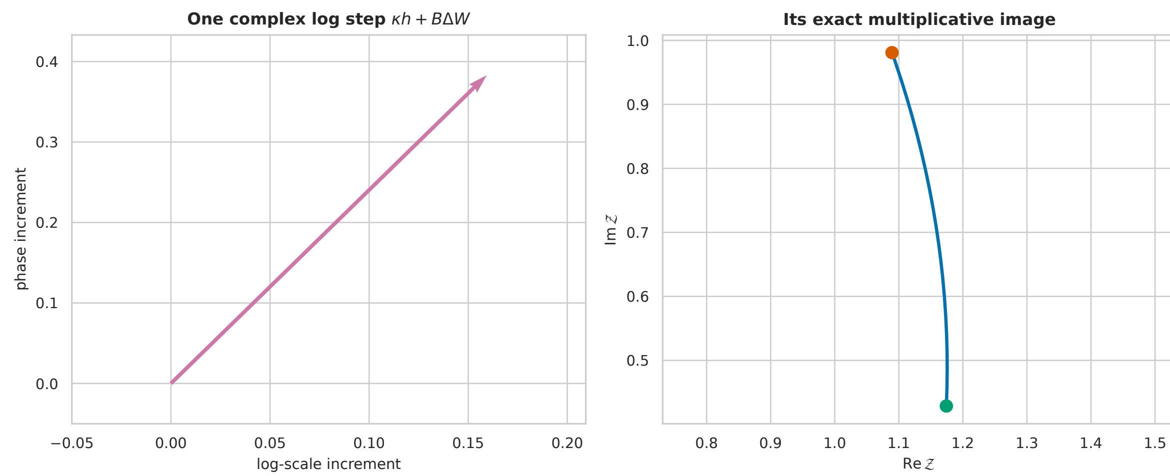

Figure 7: A complex log increment is shown as one vector whose exponential maps an initial state to the next state along a scale-and-rotation arc.

10 Transition laws and moments

10.1 Singular log-polar Gaussian law

At time t,

$$\begin{pmatrix} \log(R_t/R_0) \\ \Theta_t - \Theta_0 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} (\mu - \sigma^2/2)t \\ \omega t \end{pmatrix}, t \begin{pmatrix} \sigma^2 & \sigma\beta \\ \sigma\beta & \beta^2 \end{pmatrix} \right). \tag{13}$$

The covariance determinant is zero:

$$\sigma^2\beta^2 - (\sigma\beta)^2 = 0.$$

If $\sigma \neq 0$, the support line is

$$\Theta_t - \Theta_0 - \omega t = \frac{\beta}{\sigma}\left[\log(R_t/R_0) - \left(\mu - \frac{1}{2}\sigma^2\right) t \right]. \tag{14}$$

The simulated transition sample in Figure 8 lies on Eq. 14 up to floating-point plotting precision.

At fixed t, the complex support is parameterized by one real coordinate:

$$\left\{ \mathcal{Z}_0 e^{\kappa t + Bw} : w \in \mathbb{R} \right\}.$$

For $\sigma\beta \neq 0$, this is a logarithmic spiral. Degenerate cases become a ray ($\beta = 0$) or a circle traced through wrapped phase ($\sigma = 0, \beta \neq 0$).

Figure 9 shows why a complex-valued law need not have a two-dimensional density with respect to planar area.

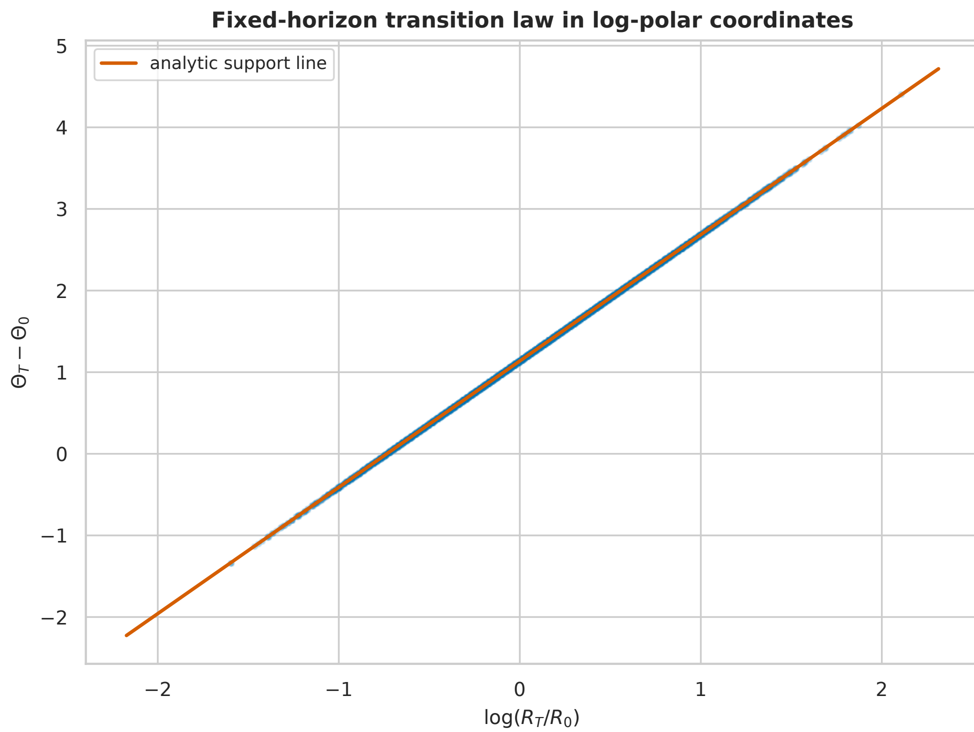

Figure 8: A fixed-horizon log-radius and phase sample lies on the analytic rank-one Gaussian support line in log-polar coordinates.

10.2 Radial moments

For real p for which the expression is considered,

$$\boxed{\mathbb{E}[R_t^p] = R_0^p \exp\left[\left(p\mu + \frac{1}{2}p(p-1)\sigma^2\right) t\right].} \tag{15}$$

All real powers have finite moments at finite t because R_t is lognormal.

10.3 Complex integer moments

For integer n,

$$\boxed{\mathbb{E}[\mathcal{Z}_t^n] = \mathcal{Z}_0^n \exp\left[\left(n\kappa + \frac{1}{2}n^2 B^2\right) t\right].} \tag{16}$$

The complex square B^2 determines both amplification or damping and rotation of the moment.

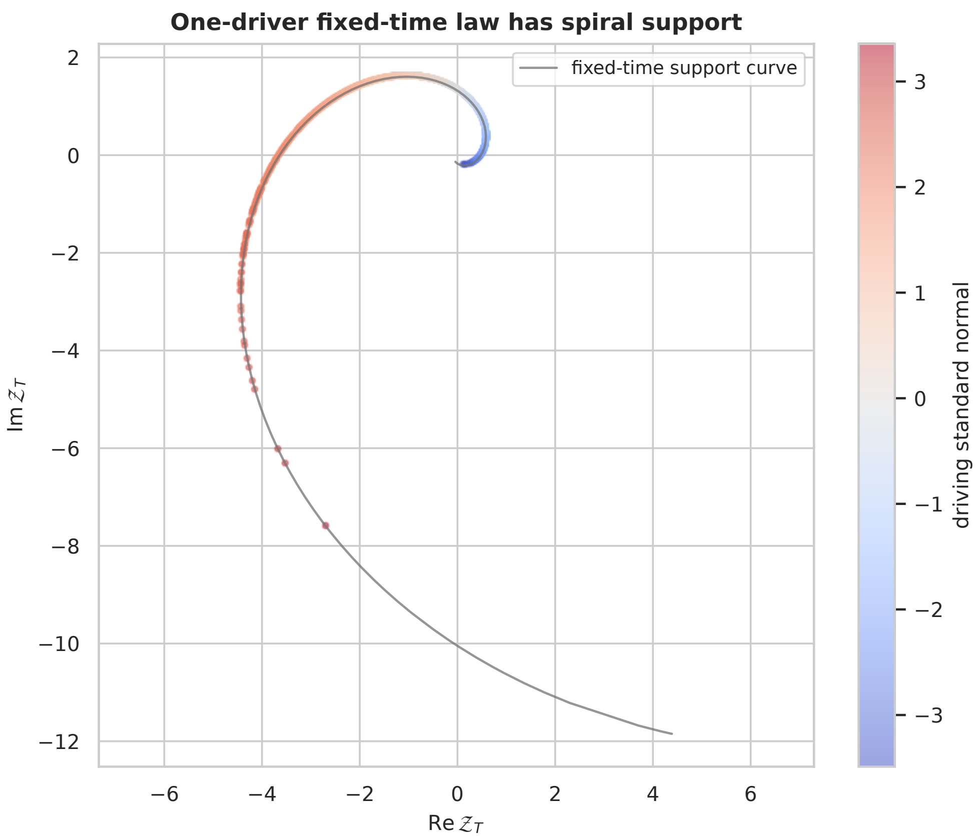

Figure 9: Terminal complex samples from one real Brownian driver lie on a one-dimensional logarithmic spiral rather than filling a two-dimensional region.

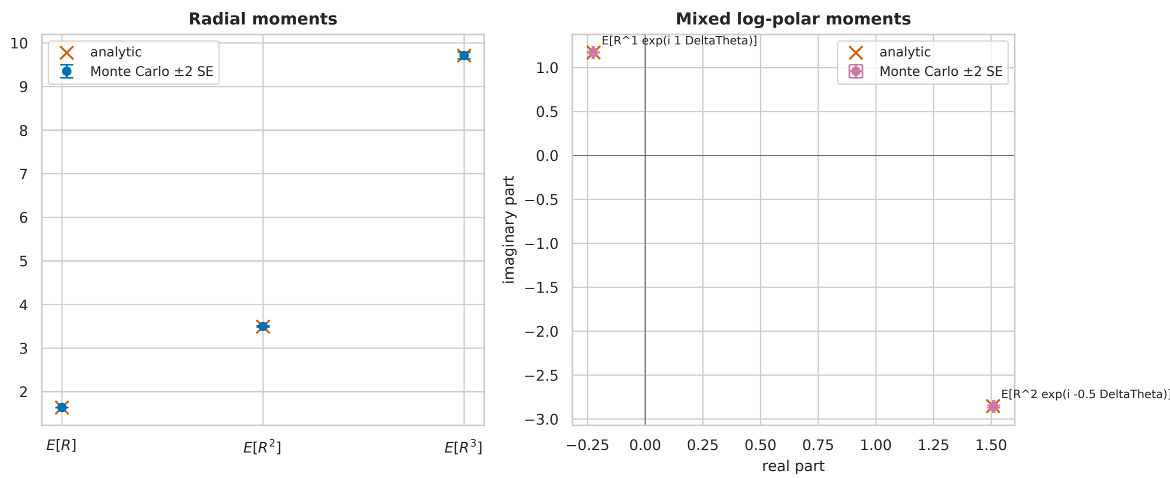

Figure 10: Analytic radial and mixed log-polar moments agree with seeded Monte Carlo estimates and their componentwise two-standard-error bars.

10.4 Mixed log-polar moments

For real p, q,

$$\boxed{\mathbb{E}\left[R_t^p e^{iq(\Theta_t-\Theta_0)}\right] = R_0^p \exp\left[p\left(\mu-\frac{1}{2}\sigma^2\right)t + iq\omega t + \frac{1}{2}(p\sigma+iq\beta)^2 t\right]}\,. \tag{17}$$

This formula exposes the shared-driver coupling through the cross term in $(p\sigma + iq\beta)^2$.

Figure 10 compares these formulas with independent terminal samples. The estimates are a numerical check; Gaussian evaluation proves the identities.

The first moment in Figure 11 need not rotate at the same rate as an individual path because the term $B^2/2$ changes both its real and imaginary exponents.

11 Geometry and noise dimension

11.1 Local Cartesian diffusion

Write $Z_t = X_t + iY_t$. This is a derived decomposition, not the definition. The complex diffusion coefficient at state z is Bz, so the real two-vector multiplying dW_t is

$$g(z) = \begin{pmatrix} \mathrm{Re}(Bz) \\ \mathrm{Im}(Bz) \end{pmatrix}.$$

The instantaneous Cartesian covariance is

$$g(z)g(z)^\mathsf{T}\,dt.$$

It is an outer product and therefore has rank at most one. Both radial and tangential components may be nonzero, but a single scalar increment chooses only one combined direction.

Figure 12 distinguishes “two geometric components” from “two independent stochastic directions.”

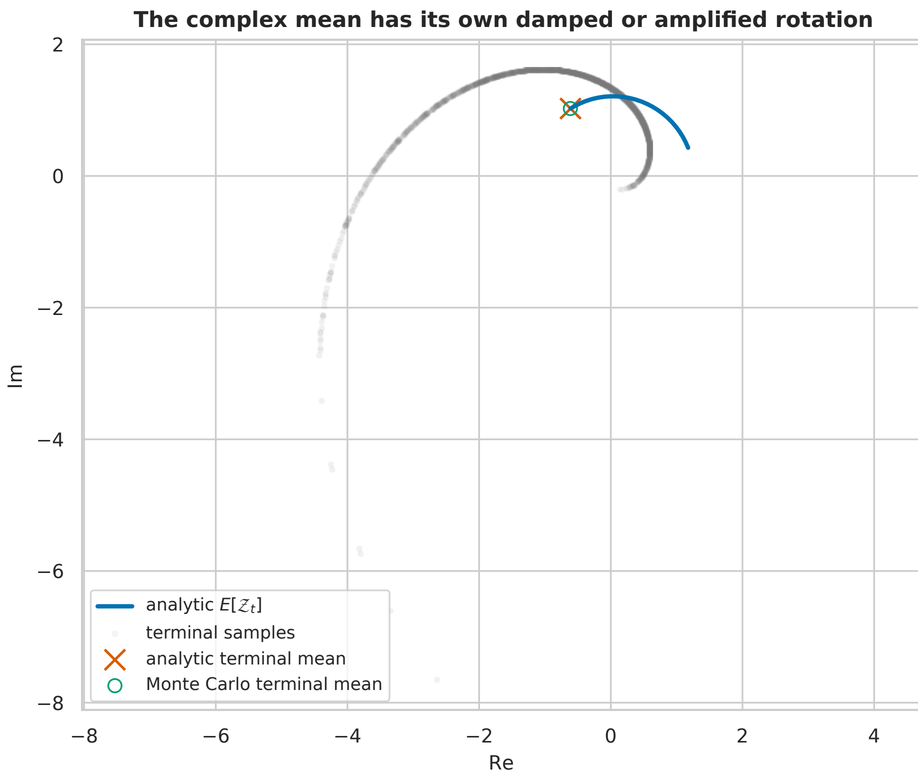

Figure 11: The analytic complex mean traces a deterministic curve through time and ends at the same point as the seeded Monte Carlo terminal mean.

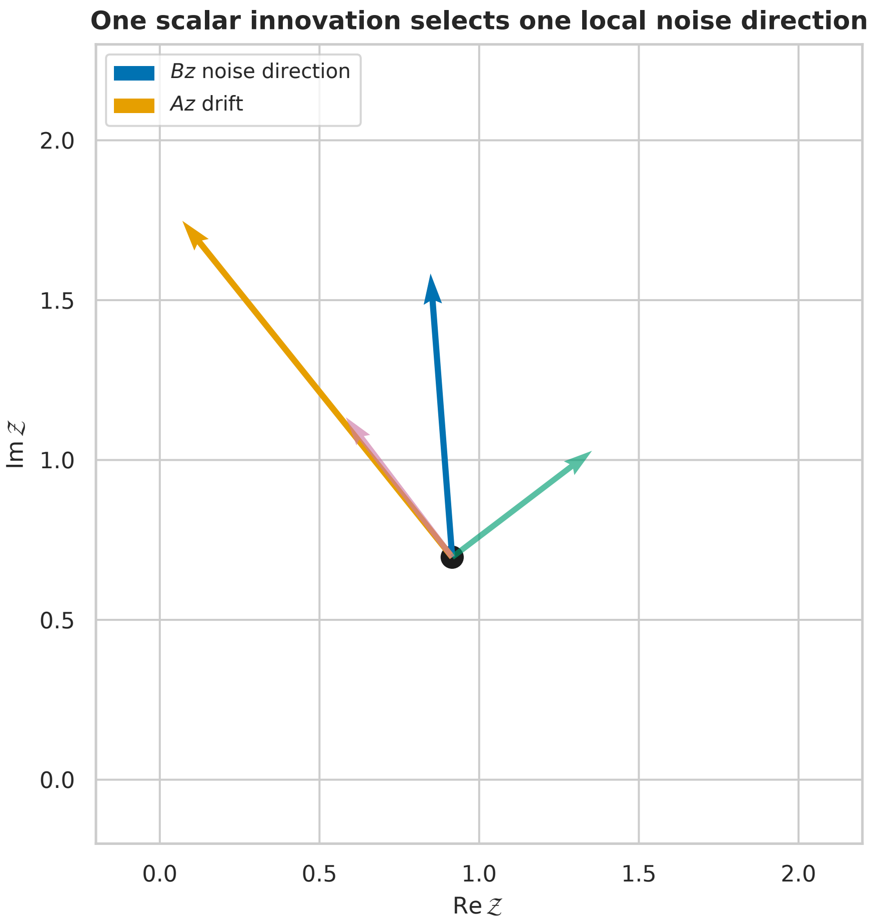

Figure 12: At one complex state, the diffusion vector has radial and tangential components but remains a single local noise direction driven by one scalar Brownian increment.

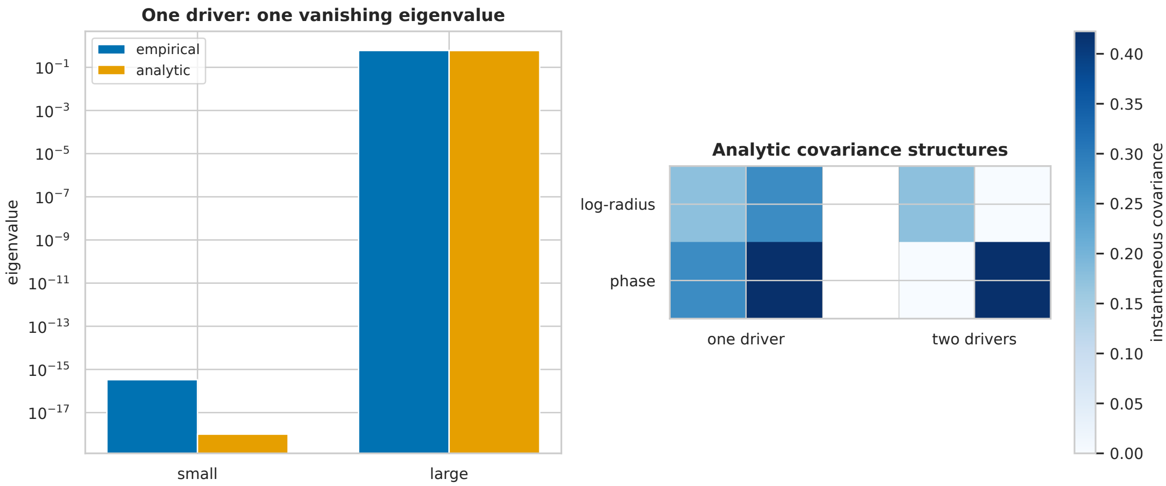

Figure 13: The one-driver covariance has one numerically vanishing eigenvalue, while a companion heatmap contrasts its outer-product structure with the two-driver diagonal covariance.

11.2 Log-polar covariance rank

The martingale parts of log-radius and phase are

$$dM_t^{(r)} = \sigma\, dW_t, \qquad dM_t^{(\theta)} = \beta\, dW_t.$$

Their covariance rate is

$$\boxed{\Sigma_1 = \begin{pmatrix} \sigma^2 & \sigma\beta \\ \sigma\beta & \beta^2 \end{pmatrix} = \begin{pmatrix} \sigma \\ \beta \end{pmatrix} \begin{pmatrix} \sigma & \beta \end{pmatrix}.} \tag{18}$$

The nonzero eigenvalue is $\sigma^2 + \beta^2$; the other is zero. This rank is invariant under any nonsingular deterministic change of the two output coordinates. Coordinate notation cannot increase the input noise dimension.

The eigenvalue ratio reported by the notebook for Figure 13 is 5.54×10^{-16}, numerical rank one.

The two clouds in Figure 14 provide the most direct visual test of the driver-count claim.

11.3 Bilinear and Hermitian brackets

From the diffusion $B\mathcal{Z}_t\, dW_t$,

$$\boxed{d[\mathcal{Z}, \mathcal{Z}]_t = B^2 \mathcal{Z}_t^2\, dt,} \tag{19}$$

and

$$\boxed{d[\mathcal{Z}, \overline{\mathcal{Z}}]_t = |B|^2 |\mathcal{Z}_t|^2\, dt.} \tag{20}$$

The first preserves complex orientation information; the second is a nonnegative accumulated energy. They answer different questions.

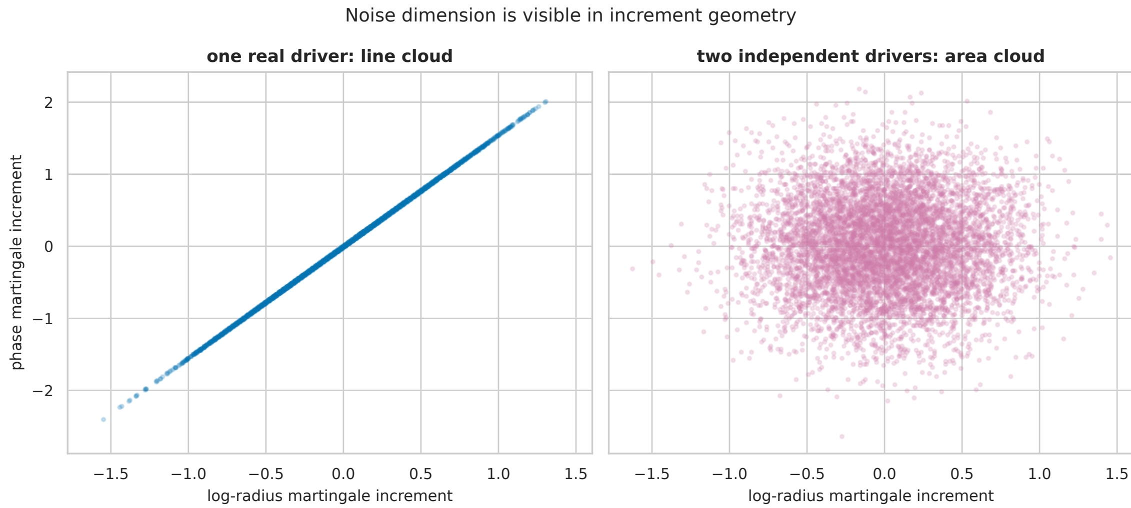

Figure 14: One-driver log-polar increments form a line, whereas independent radial and angular Brownian increments fill a two-dimensional elliptical cloud.

12 Phase 3 and two-driver boundaries

12.1 Phase 3 constrained subfamily

Phase 3 used

$$\mathcal{Z}_t = \mathcal{Z}_0 e^{(1+i\gamma)L_t},$$

where the real GBM log return is

$$L_t = \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t.$$

Expanding and matching Eq. 1 gives the necessary and sufficient restrictions

$$\boxed{\beta = \gamma\sigma, \qquad \omega = \gamma\left(\mu - \frac{1}{2}\sigma^2\right).} \tag{21}$$

Thus Phase 3 ties the angular drift and diffusion to one spiral coefficient γ. Phase 4 permits ω and β to be chosen independently while preserving the same radial GBM.

The restricted paths in Figure 15 agree to 2.73×10^{-16}, the maximum pathwise discrepancy recorded in `phase3_equivalence.json`.

12.2 Independent two-driver comparator

Let $W^{(r)}$ and $W^{(\theta)}$ be independent real Brownian motions and define

$$\mathcal{Z}_t^{(2)} = \mathcal{Z}_0 \exp\left[\kappa t + \sigma W_t^{(r)} + i\beta W_t^{(\theta)}\right]. \tag{22}$$

The modulus and phase targets remain

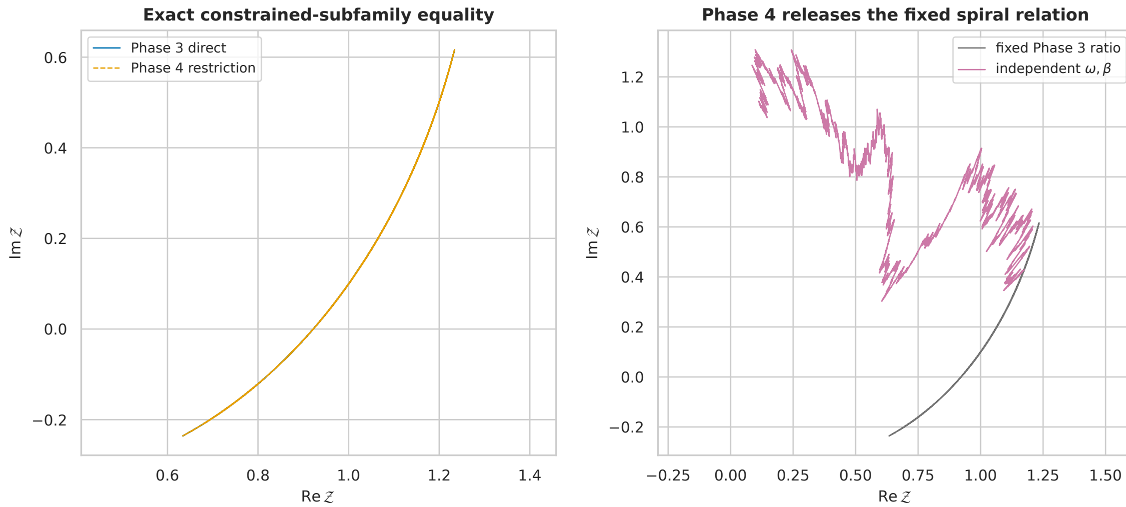

Figure 15: The Phase 3 logarithmic-spiral formula exactly overlaps the restricted Phase 4 process, while independently chosen phase parameters produce a different path.

$$\frac{dR_t^{(2)}}{R_t^{(2)}} = \mu\,dt + \sigma\,dW_t^{(r)},$$

$$d\Theta_t^{(2)} = \omega\,dt + \beta\,dW_t^{(\theta)},$$

but their covariance rate becomes

$$\boxed{\Sigma_2 = \begin{pmatrix} \sigma^2 & 0 \\ 0 & \beta^2 \end{pmatrix}.} \tag{23}$$

It has rank two when $\sigma\beta \neq 0$. The Itô bracket of the complex log increment is now

$$\sigma^2 dt + (i\beta)^2 dt = (\sigma^2 - \beta^2) dt,$$

because the cross bracket of the independent drivers vanishes. Therefore

$$\boxed{\frac{d\mathcal{Z}_t^{(2)}}{\mathcal{Z}_t^{(2)}} = \left(\mu - \frac{1}{2}\beta^2 + i\omega\right) dt + \sigma\,dW_t^{(r)} + i\beta\,dW_t^{(\theta)}.} \tag{24}$$

Compared with Eq. 10, the imaginary cross drift $\sigma\beta$ disappears. It is a signature of the one-driver coupling, not a universal feature of complex notation.

13 Numerical methods

13.1 Reproducible design

The executed discovery notebook imports the tested root `phase4_model.py`. It does not reimplement the model in an isolated paper-only module. All random number generators use seeds derived from 20260724. Figures are written as vector PDF files for the printed paper and 300-dpi PNG companions for HTML. Every evidence table is CSV or sorted JSON, and a manifest records all generated artifacts and figure alt text.

The primary numerical parameters are

$$(\mu, \sigma, \omega, \beta) = (0.18, 0.42, 0.90, 0.65),$$

with $R_0 = 1.25$, $\Theta_0 = 0.35$, and $T = 1.50$. These values are illustrative and were fixed before reporting diagnostics.

The environment record is summarized below.

Table 1: Software versions from `tables/software_versions.csv`.

Component	Version
Python	3.12.13
NumPy	2.5.1
SciPy	1.18.0
pandas	3.0.3
Matplotlib	3.11.1
Seaborn	0.13.2
SymPy	1.14.0
nbformat	5.10.4

13.2 Exact-identity checks

One Brownian path with 8192 increments tests:

- $A - B^2/2 = \kappa$;
- $|\mathcal{Z}_t| = R_t$;
- reconstruction $R_t e^{i\Theta_t} = \mathcal{Z}_t$;
- agreement of the unwrapped numerical argument with Θ_t;
- the exact one-step transition; and
- constant modulus under pure stochastic rotation.

The acceptance tolerance is 2×10^{-13}. These checks diagnose implementation consistency, while the preceding algebra supplies the proof.

13.3 Transition and covariance experiments

The transition and covariance experiment uses 250,000 seeded Gaussian increments. Empirical covariance matrices are compared with Eq. 18 and Eq. 23. Numerical rank uses a fixed 10^{-4} eigenvalue threshold; the one-driver small-to-large eigenvalue ratio must be below 10^{-12}.

13.4 Moment experiment

The moment experiment uses 600,000 independent terminal samples. For each real or complex quantity, the notebook reports componentwise Monte Carlo standard errors. A standardized residual is the empirical-minus-analytic difference divided by its estimated standard error. Every nondegenerate real and imaginary residual is required to have absolute value below 3.

13.5 Strong convergence

Euler–Maruyama is applied to Eq. 10 using 12,000 shared Brownian paths. All levels are aggregated from 512 fine increments, with step counts $16, 32, 64, 128, 256$. The terminal root-mean-square strong error is regressed against the step size on a log-log scale. The prespecified acceptance interval for the slope is $[0.42, 0.65]$, centered on the classical order 1/2 behavior described for scalar SDE simulation by Higham (2001).

13.6 Quadratic variation

A single path with 262,144 increments compares realized cumulative sums

$$\sum(\Delta\mathcal{Z})^2, \qquad \sum|\Delta\mathcal{Z}|^2$$

with the left-point integral targets in Eq. 19 and Eq. 20. Both terminal relative errors must be below 2%.

14 Numerical results

14.1 Coefficients and exactness

For the primary parameters,

Table 2: Coefficient reconciliation from `tables/coefficient_reconciliation.csv`.

Coefficient	Real part	Imaginary part
κ	0.0918	0.9000
B	0.4200	0.6500
A	-0.03125	1.1730
$A - B^2/2$	0.0918	0.9000

The coefficient residual is exactly zero at the stored double-precision values. Across the pathwise invariant suite, the largest error is 6.66×10^{-16}.

Table 3: Exact-identity diagnostics from `tables/exactness.json`.

Check	Maximum absolute error
modulus	4.44×10^{-16}
phase lift	4.44×10^{-16}
polar reconstruction	6.66×10^{-16}
exact step	1.14×10^{-16}
constant-modulus stress test	4.44×10^{-16}

14.2 Transition and covariance

At $T = 1.50$, the empirical and analytic transition summaries are:

Table 4: Fixed-horizon summaries from `tables/transition_summary.csv`.

Variable	Empirical mean	Analytic mean	Empirical variance	Analytic variance
log-radius increment	0.13681	0.13770	0.26570	0.26460
phase increment	1.34863	1.35000	0.63638	0.63375

The covariance comparison is more structurally informative.

Table 5: Empirical covariance rates from 250,000 increments in `tables/covariance_rank.csv`.

Model	$\widehat{\Sigma}_{00}$	$\widehat{\Sigma}_{01}$	$\widehat{\Sigma}_{11}$	Small eigenvalue	Large eigenvalue	Rank
one driver	0.17713	0.27413	0.42426	3.33×10^{-16}	0.60139	1
two drivers	0.17601	0.00025	0.42319	0.17601	0.42319	2

The results confirm both the rank-one theorem and the two-driver boundary. The tiny two-driver off-diagonal term is sampling error around the analytic value zero.

14.3 Moments

Selected results are shown below; the complete table includes radial orders 1–3, complex orders 1–2, and two mixed moments.

Table 6: Selected moment validation values from `tables/moment_validation.csv`; complex residual is the larger absolute componentwise residual.

Quantity	Analytic value	Monte Carlo value	Standardized residual
$\mathbb{E}[R_T]$	1.63746	1.63891	1.250
$\mathbb{E}[R_T^2]$	3.49344	3.49776	0.701
$\mathbb{E}[\mathcal{Z}_T]$	$-0.61191 + 1.02383i$	$-0.61408 + 1.02437i$	max 1.356
$\mathbb{E}[\mathcal{Z}_T^2]$	$0.31462 - 0.93185i$	$0.31935 - 0.93475i$	max 0.873

The maximum absolute standardized residual across the complete table is 1.356, comfortably inside the prespecified threshold 3.

14.4 Strong convergence

Table 7: Shared-path Euler–Maruyama data from `tables/strong_convergence.csv`.

Steps	Step size	Terminal RMS error
16	0.062500	0.21371
32	0.031250	0.13548

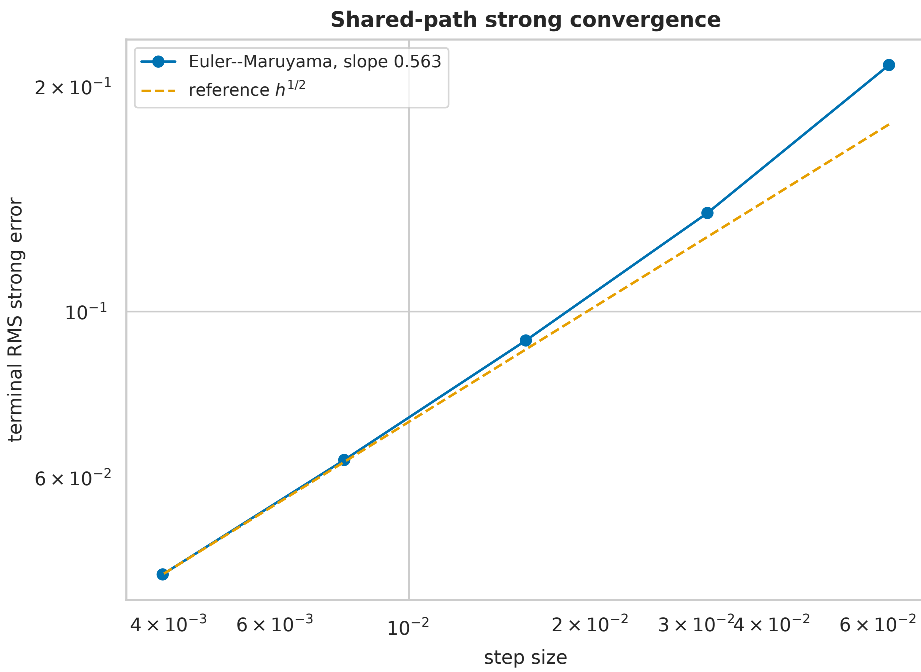

Figure 16: A log-log plot of Euler–Maruyama terminal root-mean-square error follows the reference square-root step-size line on shared Brownian paths.

Steps	Step size	Terminal RMS error
64	0.015625	0.09148
128	0.0078125	0.06331
256	0.00390625	0.04450

The fitted slope is 0.5625.

Figure 16 shows the expected order-1/2 trend. The exact transition Eq. 12 remains preferable for this constant-coefficient model; Euler–Maruyama is included as a numerical consistency check and a baseline for extensions without a closed form.

14.5 Quadratic variation

Table 8: Quadratic-variation diagnostics from `tables/quadratic_variation.csv`.

Bracket	Realized terminal value	Integral target	Relative error
bilinear	$-0.21668 - 1.42167i$	$-0.21402 - 1.42148i$	0.00186
Hermitian	1.77059	1.76728	0.00187

The simultaneous agreement in Figure 17 is a useful guard against accidentally replacing B^2 with $|B|^2$.

15 Discussion

The construction achieves the requested elimination of an external x-y state definition. The complete process is specified by one term, $\kappa t + BW_t$, inside one complex exponential. The real part of that term is the log GBM; the imaginary part is the phase lift. The corresponding exact finite-step walk has the same architecture.

The rotational role of Euler's formula is genuine. Multiplication by $e^{i\Delta\theta}$ rotates each state, and the imaginary Brownian coefficient makes $\Delta\theta$ random. What does *not* follow is an algebraic identification of $i^2 = -1$ with Brownian quadratic variation. Instead, the square of the complex diffusion coefficient enters because Itô's formula contains a quadratic-variation term. This disciplined separation preserves both the geometric

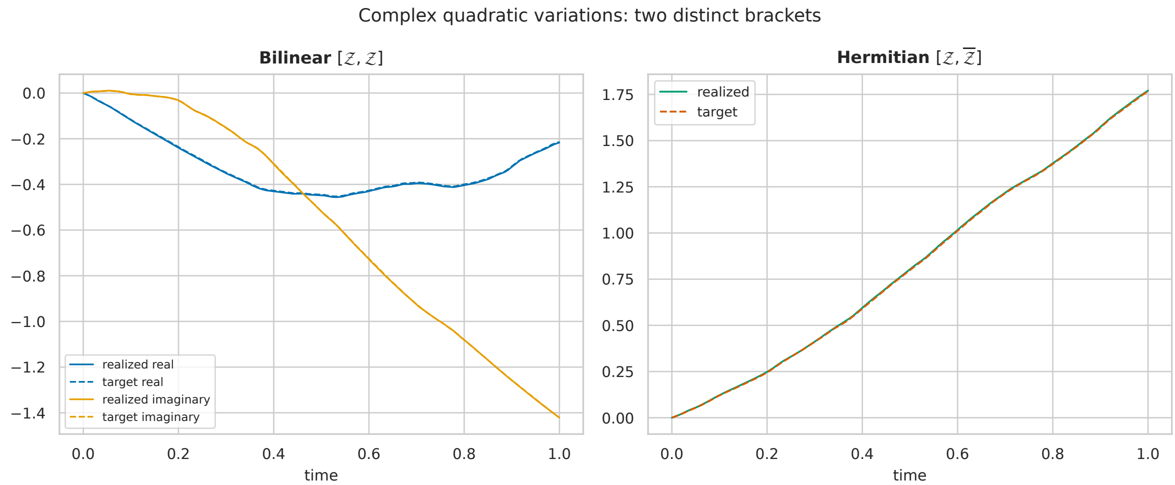

Figure 17: Cumulative realized bilinear and Hermitian complex quadratic variations track their analytic time-integral targets, with the bilinear bracket split into real and imaginary parts.

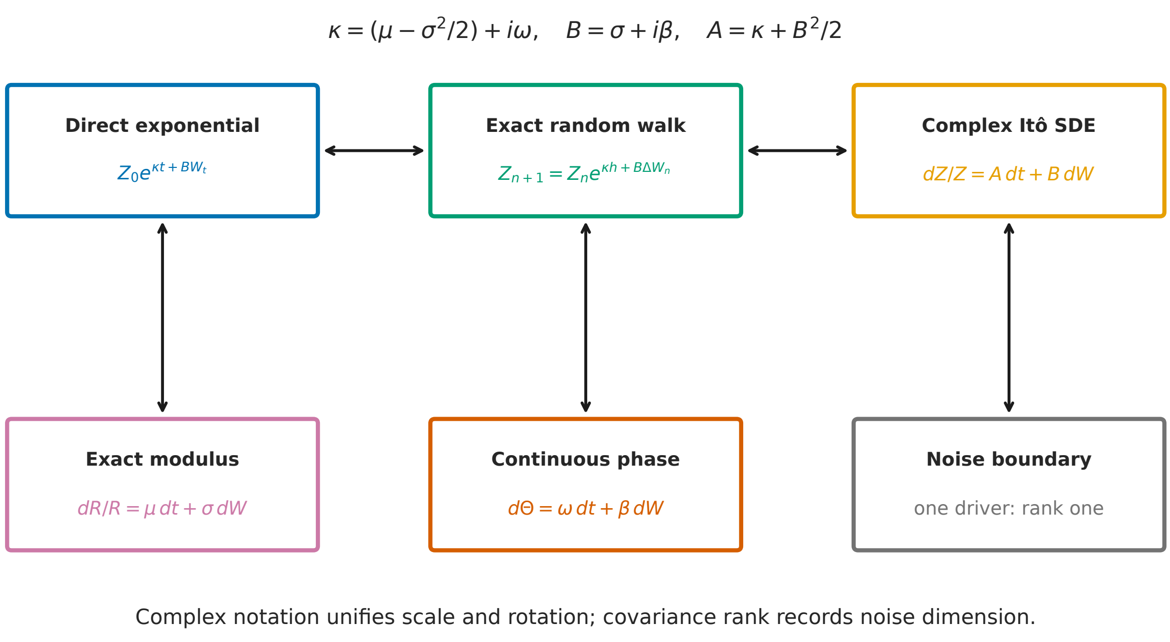

Figure 18: A taxonomy links the direct exponential, exact random walk, complex Itô SDE, exact GBM modulus, continuous phase, and rank-one noise boundary.

3. **Singular fixed-time support.** For one driver, the joint log-polar law is singular and the complex fixed-time support is one-dimensional.
4. **Phase conventions.** The continuous phase requires a chosen initial lift. The principal argument remains discontinuous at its branch cut.
5. **No zeros.** The exponential construction stays in $\mathbb{C} \setminus \{0\}$. It does not model paths that hit the origin.
6. **Constant coefficients.** Exact transition and closed-form moments rely on constant coefficients. State-dependent or noncommuting matrix extensions require additional analysis.
7. **No planar Brownian claim.** Classical planar Brownian motion and its winding laws (Spitzer 1958; Pitman and Yor 1986) are not laws of this one-driver process.
8. **No financial assertion.** Although GBM terminology is standard in mathematical finance, this paper makes no pricing, calibration, or no-arbitrage claim.

17 Conclusion

The design objective admits one concise native complex form:

$$\boxed{\mathcal{Z}_t = \mathcal{Z}_0 \exp\left(\left[\left(\mu - \frac{1}{2}\sigma^2 \right) + i\omega \right] t + (\sigma + i\beta) W_t \right).}$$

Its exact random-walk transition is

$$\boxed{\mathcal{Z}_{n+1} = \mathcal{Z}_n \exp\left(\left[\left(\mu - \frac{1}{2}\sigma^2 \right) + i\omega \right] h + (\sigma + i\beta) \Delta W_n \right).}$$

Its equivalent complex Itô SDE is

$$\boxed{\frac{d\mathcal{Z}_t}{\mathcal{Z}_t} = \left[\mu - \frac{1}{2}\beta^2 + i(\omega + \sigma\beta) \right] dt + (\sigma + i\beta) dW_t.}$$

The modulus is exactly the desired real GBM, and the imaginary exponent provides deterministic and stochastic rotation. The cross drift and rank-one covariance are not incidental: they precisely record the use of one shared real Brownian driver. In this way, complex notation supplies a unified scale–rotation language while covariance rank keeps the stochastic dimension honest.

18 Appendix A: Expanded Itô algebra

Write

$$\Lambda_t = U_t + iV_t,$$

where

$$dU_t = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma \, dW_t,$$

$$dV_t = \omega \, dt + \beta \, dW_t.$$

In quadratic-covariation notation, the two real processes have Itô products

$$(dU_t)^2 = \sigma^2 dt, \qquad (dV_t)^2 = \beta^2 dt, \qquad dU_t\, dV_t = \sigma\beta\, dt.$$

Since

$$\mathcal{Z}_t = \mathcal{Z}_0 e^{U_t}(\cos V_t + i \sin V_t),$$

one may apply the two-dimensional real Itô formula. More compactly,

$$(d\Lambda_t)^2 = (dU_t + i\, dV_t)^2 = (dU_t)^2 - (dV_t)^2 + 2i\, dU_t\, dV_t,$$

so

$$(d\Lambda_t)^2 = (\sigma^2 - \beta^2 + 2i\sigma\beta)dt = B^2 dt.$$

Then

$$\frac{d\mathcal{Z}_t}{\mathcal{Z}_t} = d\Lambda_t + \frac{1}{2}(d\Lambda_t)^2,$$

which expands to Eq. 10. This calculation makes clear that the imaginary cross drift arises from $dU_t\, dV_t$, the shared Brownian covariation.

19 Appendix B: Moment derivations

For any complex c,

$$\mathbb{E}[e^{cW_t}] = e^{c^2 t/2},$$

by analytic continuation of the Gaussian moment-generating function. For the radial moment, substitute

$$R_t^p = R_0^p e^{p(\mu - \sigma^2/2)t + p\sigma W_t}$$

and use $c = p\sigma$. This yields Eq. 15.

For the complex moment,

$$\mathcal{Z}_t^n = \mathcal{Z}_0^n e^{n\kappa t + nBW_t},$$

and $c = nB$ yields Eq. 16.

Finally,

$$R_t^p e^{iq(\Theta_t - \Theta_0)} = R_0^p \exp\left[p\left(\mu - \frac{1}{2}\sigma^2\right)t + iq\omega t + (p\sigma + iq\beta)\, W_t\right],$$

which gives Eq. 17.

20 Appendix C: Reproducibility manifest

The canonical source is `index.qmd`. The discovery record is `notebooks/phase4_discovery.ipynb`, generated by `notebooks/build_phase4_discovery.py`. The builder writes:

- 18 vector figures under `figures/vector/`;
- 18 raster companions under `figures/raster/`;
- coefficient, exactness, transition, covariance, moment, Phase 3, convergence, bracket, and software evidence under `tables/`; and
- figure and table manifests with stable relative paths.

Two clean notebook executions completed with sequential code-cell counts 1–9, no error outputs, identical code sources, identical deterministic stream text, and the terminal marker `ALL_PHASE4_ASSERTIONS_PASSED`. The generated table hashes are recorded in the project reproducibility guide.

The paper uses simulations to check implementation and illustrate geometry. All the central identities—modulus, phase, Itô drift, exact transition, moments, brackets, covariance rank, Phase 3 restrictions, and two-driver drift—are analytical.

Abdulle, Assyr, Gilles Vilmart, and Konstantinos C. Zygalakis. 2013. "Mean-Square a-Stable Diagonally Drift-Implicit Integrators of Weak Second Order for Stiff Itô Stochastic Differential Equations." *BIT Numerical Mathematics* 53 (4): 827–40. https://doi.org/10.1007/s10543-013-0430-8.

Buckwar, Evelyn, Rózsa Horváth-Bokor, and Renate Winkler. 2006. "Asymptotic Mean-Square Stability of Two-Step Methods for Stochastic Ordinary Differential Equations." *BIT Numerical Mathematics* 46 (2): 261–82. https://doi.org/10.1007/s10543-006-0060-5.

Černý, Aleš, and Johannes Ruf. 2022. "Simplified Stochastic Calculus via Semimartingale Representations." *Electronic Journal of Probability* 27 (3): 1–32. https://doi.org/10.1214/21-EJP729.

Černý, Aleš, and Johannes Ruf. 2023. "Simplified Calculus for Semimartingales: Multiplicative Compensators and Changes of Measure." *Stochastic Processes and Their Applications* 161: 572–602. https://doi.org/10.1016/j.spa.2023.04.010.

Doléans-Dade, Catherine. 1970. "Quelques Applications de La Formule de Changement de Variables Pour Les Semimartingales." *Zeitschrift für Wahrscheinlichkeitstheorie Und Verwandte Gebiete* 16: 181–94.

Higham, Desmond J. 2001. "An Algorithmic Introduction to Numerical Simulation of Stochastic Differential Equations." *SIAM Review* 43 (3): 525–46. https://doi.org/10.1137/S0036144500378302.

Itô, Kiyosi. 1951. "On a Formula Concerning Stochastic Differentials." *Nagoya Mathematical Journal* 3: 55–65. https://doi.org/10.1017/S0027763000012216.

Larsson, Martin, and Johannes Ruf. 2019. "Stochastic Exponentials and Logarithms on Stochastic Intervals—a Survey." *Journal of Mathematical Analysis and Applications* 476 (1): 2–12. https://doi.org/10.1016/j.jmaa.2018.11.040.

Mora, Carlos M., Juan C. Jimenez, and Hector Selva. 2015. "A Stable Numerical Scheme for Stochastic Differential Equations with Multiplicative Noise." *SIAM Journal on Numerical Analysis* 53 (4): 1613–41. https://doi.org/10.1137/140984488.

Mörters, Peter, and Yuval Peres. 2010. *Brownian Motion*. Cambridge Series in Statistical and Probabilistic Mathematics 30. Cambridge University Press. https://doi.org/10.1017/CBO9780511750489.

Perret, Christian. 2010. "The Stability of Numerical Simulations of Complex Stochastic Differential Equations." Doctoral Dissertation No. 18868. ETH Zürich. https://doi.org/10.3929/ethz-a-006026507.

Pitman, Jim, and Marc Yor. 1986. "Asymptotic Laws of Planar Brownian Motion." *The Annals of Probability* 14 (3): 733–79. https://doi.org/10.1214/aop/1176992436.

Spitzer, Frank. 1958. "Some Theorems Concerning 2-Dimensional Brownian Motion." *Transactions of the American Mathematical Society* 87 (1): 187–97. https://doi.org/10.1090/S0002-9947-1958-0104296-5.